

ADAM-TPU: A Sovereign Complex-Spectral Lattice for Cognitively-Immune LLM Inference and Training via Radial Coalescence

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Abstract

This paper presents an architectural proposal for ADAM-TPU, a non-von Neumann tensorial core for Large Language Model (LLM) inference and constrained on-device adaptation, built on an analog memristive-photonic substrate operating natively in the complex field $\mathbb{C}$. Conventional accelerators amortize the matrix-vector multiply (MVM) at the heart of transformer workloads across large digital multiply-accumulate (MAC) arrays, repeatedly paying data-movement and conversion costs. We propose instead to *deploy* low-rank weight factorizations directly onto an analog crossbar, where a single physical settling step performs the MVM via Kirchhoff/Ohm summation, and to realize the orthogonal/unitary factors optically using established interferometric meshes. Because coherent photonics carries magnitude *and* phase, the substrate is complex-valued by construction: we encode embeddings as $z = re^{i\phi}$, realize complex weights with paired conductance cells and a $\pi/2$ phase element, and exploit the spectral theorem so that Hermitian similarity operators retain real spectra. We pair this with (i) a VLIW/EPIC front-end that moves scheduling into the compiler and treats the compiler as a conductor of physical actuators; (ii) a CORDIC radial datapath aligned with the directional geometry of normalized embeddings; (iii) a context layer that connects a recurrent low-rank state (linear-attention/state-space) with the rotation-based KV-cache quantization frontier, observing that a photonic unitary mesh can *physically* supply the random rotation those methods perform in software; and (iv) an execution-integrity layer based on measured boot, control-flow integrity, and an *optical* physical unclonable function. A two-plane design pairs an open, attestable host control-plane with this analog data-plane, and a *hardware-aware distillation* pipeline trains

models directly into the substrate’s low-rank, rotated, quantized, noisy regime—so the chip need only *apply* a factorization, never compute one—while repurposing device noise as sampling temperature and physical decay as recurrent state, each with stated bounds. We are explicit about what is demonstrated in the literature, what is plausible extrapolation, and what is open. **This is a position and architecture paper, not a fabricated-and-measured result.** Released under the S.V.E. Meta-License v4.0 (share-alike); its falsifiable open problems (§12) are part of the contribution.

Keywords: In-Memory Computing, Complex-Valued Networks, Photonic Unitary Meshes, Low-Rank Approximation, Rotary/Spherical Embeddings, KV-Cache Quantization, Optical PUF, Hardware Attestation, S.V.E. Meta-License v4.0.

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1 Status and Scope

To keep the reader correctly calibrated: ADAM-TPU has *not* been fabricated, taped out, or benchmarked. No quantitative speedup, energy, or accuracy figure attributed to ADAM-TPU in this document is a measured result; numbers that do appear are either drawn from the cited literature (and attributed there) or stated as targets. The purpose is to (a) assemble individually-demonstrated primitives into a coherent architecture, (b) separate cleanly the parts supported by existing experiments from extrapolation, and (c) state the open problems precisely enough to be attacked or refuted. A preprint that conflates “demonstrated” with “imagined” is the failure mode the S.V.E. framework exists to prevent.

2 Introduction

Transformer workloads are dominated by dense MVM. State-of-the-art digital accelerators such as the TPU implement this as a systolic array of MAC units [1]; they are already *dataflow* machines and spend comparatively little area on the speculative out-of-order (OoO) machinery that dominates general-purpose CPUs. The pressure point in such accelerators is therefore not control logic but the *memory wall* and, in the analog case, the *conversion wall*.

We propose to attack the MVM at the level of physics. An analog resistive crossbar computes a full MVM in one settling step by Ohm’s and Kirchhoff’s laws [2, 3, 4]; coherent photonic meshes implement arbitrary unitary transforms by construction [6, 7, 8]. Because optics is intrinsically a wave phenomenon, the natural arithmetic of such a substrate is complex-valued, which aligns with a mature line of complex- and unitary-valued neural networks [9, 10]. ADAM-TPU combines these substrates with three software-aligned ideas that already work in the digital world—low-rank factorization, rotary/spherical embedding geometry,

and modern context compression—and an execution-integrity layer assembled from standard hardware-security primitives, including an optical PUF [23].

Terminology. We use loaded terms operationally, not rhetorically. *Sovereign* = supply-chain-attestable and vendor-independent (the running configuration can be verified against a published specification). *Cognitively immune* = execution whose loaded configuration is cryptographically attestable and whose control flow cannot be silently overridden by undocumented microcode. It is a property of the trust chain, not a claim about model cognition. *Radial coalescence* = computing attention/similarity as phase alignment on the unit sphere rather than as Cartesian inner products.

Contributions.

1. A clean distinction between *applying* a pre-computed factorization on a crossbar (well supported) and *solving* a decomposition physically in a closed analog loop (supported only at small scale [5]); the datapath relies on the former.
2. A *complex-valued* core: photonic phase as the imaginary axis, paired memristor cells for Re/Im conductance, and the spectral theorem for Hermitian operators—with an honest scope on where the Hermitian-real-spectrum property actually applies (§4.3).
3. A radial/CORDIC datapath [11, 12] motivated by the geometry that already makes rotary position embeddings effective [13].
4. A context layer linking the recurrent low-rank state [14, 15, 16] to the rotation-based KV-quantization frontier [17, 18, 19, 20], with the observation that a photonic mesh can physically supply the random rotation these methods use in software (§7.2).
5. An execution-integrity layer built from *feasible* primitives—measured boot [27], control-flow integrity [26], and an optical PUF [23]—rather than the impossible task of pre-registering every transient runtime state.
6. A two-plane system design—an open, attestable RISC-V/EPIC host control-plane plus the analog data-plane—driven by a single wide instruction word spanning both, with an honest treatment of digital versus *physical* side channels (§5).
7. A hardware-aware distillation pipeline that trains the model into the substrate’s low-rank/rotated/quantized/noisy regime—*resolving the apply $\neq$ solve risk*—and that repurposes device noise as sampling temperature and physical decay as recurrent state, each with stated bounds (§8).
8. An explicit, falsifiable limitations and improvements section (§12).

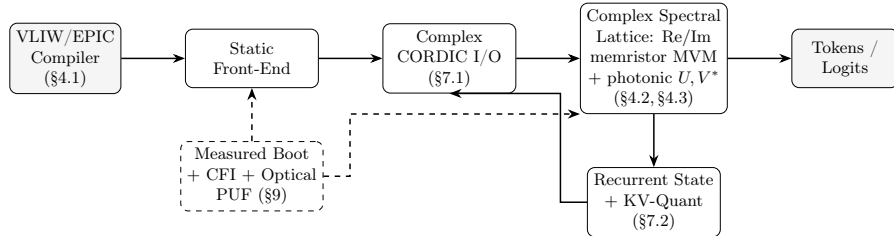


Figure 1: ADAM-TPU at a glance. Scheduling is resolved at compile time; the complex-valued spectral lattice performs the MVM and unitary rotations; a recurrent state plus KV-quantization replace an unbounded cache; an out-of-band integrity layer (dashed) attests the configuration, guards control flow, and roots identity in an optical PUF. Dashed boxes are the most speculative.

3 Related Work and Positioning

Analog in-memory MVM. Resistive crossbars performing MVM in the analog domain are well established [2, 3, 4], as are the constraints we inherit: limited bit-precision, device variability, and peripheral ADC/DAC energy dominance.

Physical solving vs. applying. A feedforward crossbar *applies* a known matrix; it does not by itself *compute* that matrix’s SVD. Closed-loop crossbar circuits that solve linear systems and eigenproblems in one step exist [5], but at small dimension; physical decomposition at LLM scale is an open target (§12).

Photonic and complex/unitary computing. Any unitary factors into a mesh of Mach–Zehnder interferometers and phase shifters [7, 8], used for optical neural inference [6]. Complex- and unitary-valued networks have an independent algorithmic history [9, 10]; we connect the two.

Radial geometry. Normalized embeddings live on a sphere, and rotation in embedding space is the operating principle of RoPE [13]; CORDIC [11, 12] is the canonical shift-add hardware for rotations and for rectangular↔polar conversion.

Context: state vs. bits. Two complementary families reduce KV cost. (i) *State* methods replace the growing cache with a fixed-size recurrent state [14, 15, 16]. (ii) *Quantization* methods reduce bits per entry: TurboQuant [17] randomly rotates vectors so coordinates become near-independent and then applies optimal scalar quantization plus a 1-bit QJL residual [18], reaching near-Shannon distortion and reporting quality-neutral KV at 3.5 bits; PolarQuant [19] quantizes via a polar transformation; QuaRot/RotateKV [20, 21] exploit rotation to remove outliers before low-bit inference. Their reliance on a random rotation makes them, surprisingly, a natural fit for a photonic substrate (§7.2).

Hardware integrity. Measured boot and remote attestation [27], control-flow integrity [26], electronic PUFs [24, 25], and optical/scattering PUFs [23] are mature; we compose them.

4 Foundational Architecture: VLIW + Spectral Lattice

4.1 Compiler as Conductor (VLIW/EPIC)

General-purpose OoO CPUs spend large area and power on scheduling, branch prediction, and speculation. Explicitly-parallel designs (EPIC [22]; the VLIW Elbrus line) move that work into the compiler, and dataflow accelerators such as the TPU already avoid most OoO logic [1]. ADAM-TPU makes the schedule fully static: a Very Long Instruction Word resolves all dependencies, routing, and lattice configuration at compile time. Crucially, the word controls not only arithmetic but *physical actuators*—laser-diode drive (input phases), DAC settings, and the switch matrix that gates spectral rank (§4.4). The compiler thus conducts on-chip physics. We do *not* claim a fixed “40%” area reclaim—that figure describes OoO CPUs, not dataflow arrays; the realistic gain over an already-systolic baseline is modest and is paid for in compiler complexity (§12).

4.2 Analog Substrate: Apply, then (perhaps) Solve

Let $W \in \mathbb{R}^{M \times N}$ be a weight matrix mapped onto conductances G_{ij} of a non-volatile crossbar; inputs are encoded as voltages V_i (or optical amplitudes). Column currents realize the MVM in one settling step:

$$I_j = \sum_{i=1}^M V_i G_{ij}. \quad (1)$$

We deploy W in factored form. Its (offline-computed) SVD is

$$W = U \Sigma V^* = \sum_{k=1}^r \sigma_k u_k v_k^*, \quad (2)$$

where the unitary factors U, V^* are realized by photonic MZI meshes [7, 8, 6] and the nonnegative real diagonal Σ by the crossbar. The lattice thus *applies* a low-rank transform in constant settling time—constant in latency, not in resources: the array is $O(MN)$ in devices and readout is $O(M)$ in ADC conversions, where most energy goes (§12). Computing (2) in place via a closed analog loop [5] is a separate, small-scale capability flagged as a research target, not a datapath assumption.

4.3 Complex-Valued Core and the Hermitian Trick

Coherent optics carries magnitude and phase, so the substrate is complex-valued for free. We encode each embedding component as

$$z_k = r_k e^{i\phi_k} = r_k (\cos \phi_k + i \sin \phi_k), \quad (3)$$

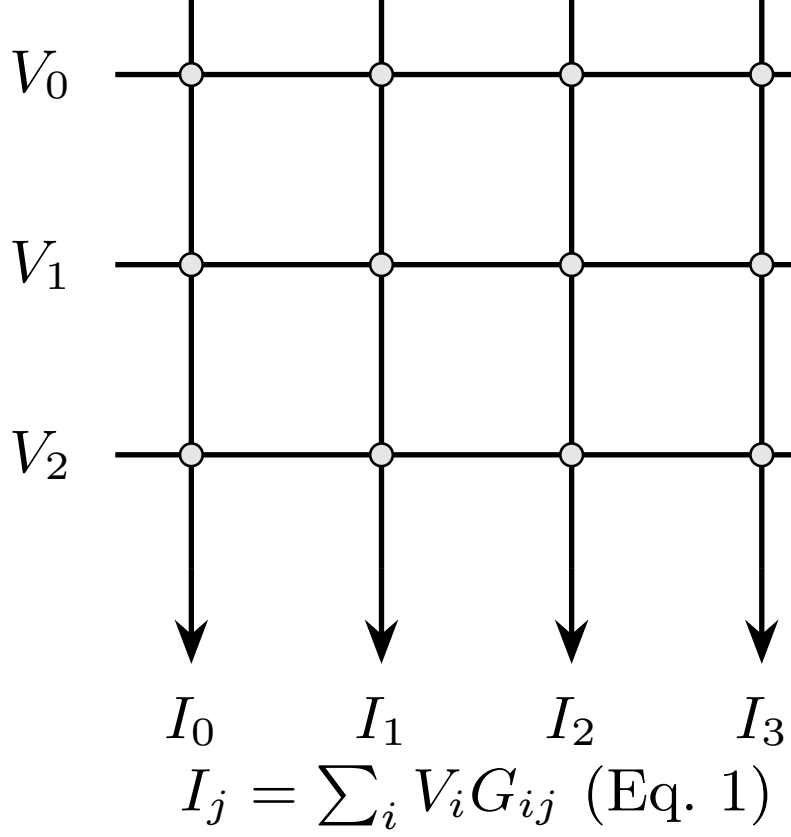


Figure 2: Analog crossbar MVM: rows drive voltages V_i , cross-points hold conductances G_{ij} , column currents I_j realize dot products by Kirchhoff summation in one settling step.

with r_k a magnitude (salience) and ϕ_k a phase (positional/semantic angle). A complex weight is realized by a *paired* conductance cell plus a fixed $\pi/2$ phase element:

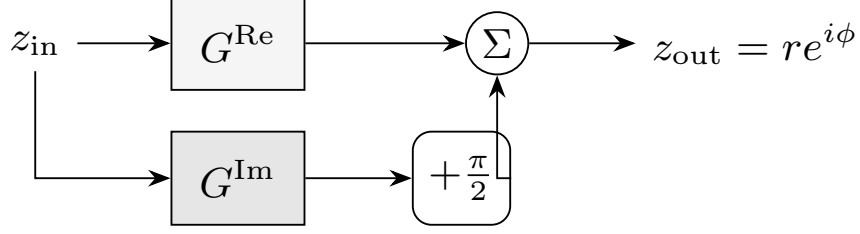
$$G_k = G_k^{\text{Re}} + i G_k^{\text{Im}}. \quad (4)$$

In polar form multiplication is magnitude product and *phase addition*,

$$z_1 z_2 = (r_1 r_2) e^{i(\phi_1 + \phi_2)}, \quad (5)$$

so phase shifters perform the angular part while coherent detection reads the interference encoding the complex inner product. This lowers (but does not eliminate—see §12) intermediate conversion.

Where Hermitian structure helps—and where it does not. For a *Hermitian* operator $H = H^*$, the spectral theorem guarantees real eigenvalues and



paired memristor cell + photonic $\frac{\pi}{2}$ shift
realize $G = G^{\text{Re}} + iG^{\text{Im}}$ (Eq. 4)

Figure 3: Complex MVM element. A real and an imaginary conductance cell, combined through a $\pi/2$ phase element, implement a complex weight; multiplication becomes magnitude product and phase addition (Eq. 5).

an orthonormal eigenbasis:

$$H = \sum_{n=1}^R \lambda_n u_n u_n^*, \quad \lambda_n \in \mathbb{R}. \quad (6)$$

This is genuinely useful for the *attention similarity operator* and other normal/Gram operators (e.g. XX^*), whose spectra are naturally real and whose eigenvectors a unitary mesh can apply. It does *not* apply to an arbitrary weight matrix: a general $W \in \mathbb{R}^{M \times N}$ is neither square nor Hermitian. For general W the honest tool is the SVD of (2)—two unitary meshes and a real diagonal—not a Hermitian eigendecomposition. We therefore use (6) only for symmetric operators and (2) elsewhere; conflating the two is a common and avoidable error.

4.4 Dynamic Truncation and Scalable Precision

Because peripheral conversion dominates energy, rank is a runtime knob. With $\sigma_1 \geq \dots \geq \sigma_r$, the rank- K approximation

$$W_K = \sum_{k=1}^K \sigma_k u_k v_k^* \quad (7)$$

is optimal in Frobenius and spectral norm (Eckart–Young). A switch matrix gates the spectral tail $k > K$: low K for cheap inference, full rank $k \rightarrow r$ when the Cognitive Operating System (CogOS) requests a high-precision Systemic Verification Engineering (SVE) audit pass.

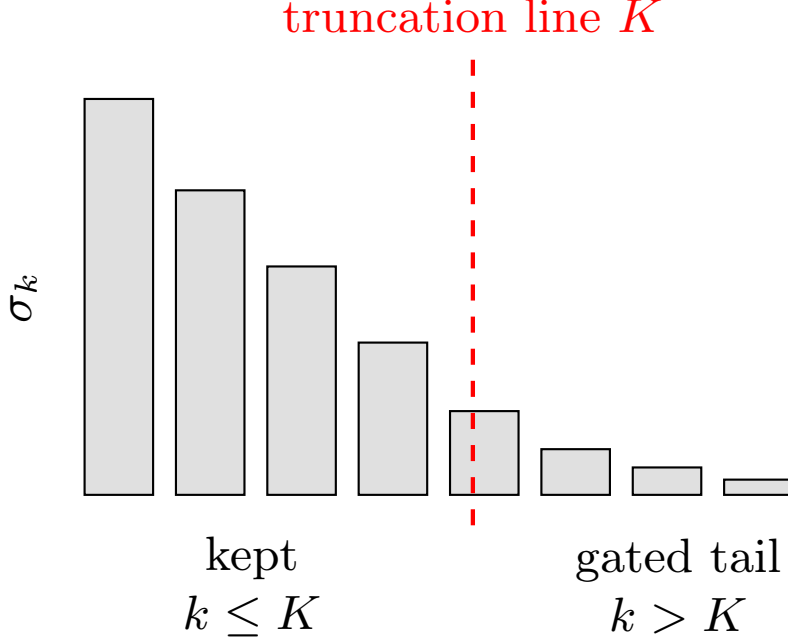


Figure 4: Dynamic truncation. Singular values are sorted; a switch gates the low-energy tail for cheap inference and reconnects it for full-precision audit (Eq. 7).

5 System Architecture: Host Control-Plane and ADAM Data-Plane

ADAM-TPU does not operate in a vacuum; it is the data-plane of a two-tier system. A general-purpose, open, and verifiable *host* (control-plane) handles what analog tensor hardware cannot: the operating system and CogOS launch, tokenization/detokenization, file and peripheral I/O, and orchestration. The host is a clean RISC-style core—auditable RISC-V, or an explicitly-parallel VLIW design in the EPIC/Elbrus lineage [22]—chosen so the control-plane is itself attestable (§9). The ADAM-TPU data-plane executes the heavy transformer kernels: MVM, rotary embedding, sparse attention, and context state.

The compiler issues one wide word across both planes. The VLIW/EPIC compiler (§4.1) emits, per issue, a single wide instruction word spanning the system,

$$w_t = \left(\underbrace{c_t}_{\text{host}}, \underbrace{\{\theta_m\}}_{\text{MZI phases}}, \underbrace{K_t}_{\text{rank}} \right), \quad (8)$$

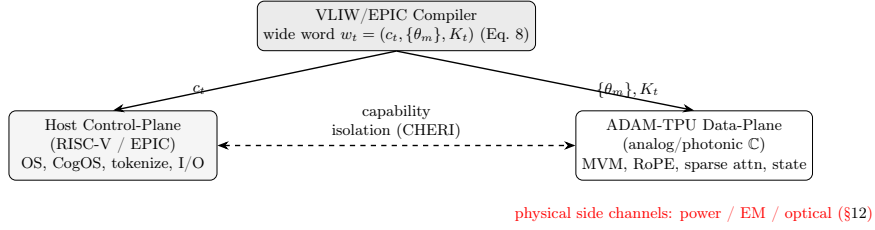


Figure 5: Two-plane system. An open, attestable host runs the OS/CogOS and I/O; the analog data-plane runs the tensor kernels. One wide instruction word (Eq. 8) configures both planes, the mesh phases, and the rank switch. Digital isolation does not cover physical side channels.

where c_t drives host-side digital control, $\{\theta_m\}$ configures the phase-shifter mesh for the unitary rotation of §7.2 (random *or* learned, §8), and K_t sets the analog switch matrix gating spectral rank as a function of token entropy. The compiler thus schedules physics across both planes, not merely instructions.

Isolation, and an honest note on side channels. Host and data-plane are physically separated, with capability-based compartmentalization between execution domains and the analog routing matrix; hardware capabilities (CHERI [37]) are the established mechanism for such memory-safety and confused-deputy mitigation, and a non-shared analog datapath removes much of the shared-microarchitecture surface that speculative attacks exploit [38]. We do not overclaim: this mitigates *classes* of attacks; it does not “eliminate” timing side channels in general. Worse, an analog/photonic data-plane introduces *physical* side channels of its own—power, electromagnetic, and optical emission—and the optical-PUF probe (§9.3) is itself a leakage surface. Physical side-channel resistance is therefore a separate requirement (§12), not a corollary of digital isolation.

6 Two-Chip Co-Design: ADAM Control-Plane and ADAM-TPU Data-Plane

The architecture of §5 is naturally realized as two physical dies: *ADAM*, a digital general-purpose control-plane (auditable RISC-V or an EPIC/VLIW core), and *ADAM-TPU*, the analog photonic–memristive data-plane. This is textbook control-plane/data-plane co-design: discrete logic where bit-exact determinism is required, wave transforms where geometry and throughput dominate. The host owns the operating system and CogOS, memory and I/O, and—critically—tokenization: text cannot enter an analog array directly, so ADAM converts tokens to indices and then to laser/DAC drive. The wide instruction word of Eq. 8 keeps the two dies in lock-step, issuing host control and mesh/rank configuration in the same cycle.

6.1 The Boundary Is the Bottleneck

The hard problem of a two-die system is not the cores but the link between them. If activations are to stay in the analog/optical domain across layers (§6.2), the die-to-die boundary must carry analog or optical signals; prevailing chiplet interconnect standards are electrical and digital, so an analog-preserving boundary requires co-packaged optics or optical I/O between dies—an active research area and, on this design, the dominant integration risk. A 2.5D interposer or 3D stack shortens the distance, but the signalling discipline at the seam, not the floorplan, decides whether the analog dataflow survives the crossing.

6.2 Deferred Conversion, Stated Honestly

The energy argument for the split is that intermediate results need not return to the digital domain at every layer: keeping phase/amplitude in the optical path across consecutive linear stages cuts ADC/DAC traffic, the dominant peripheral cost (§12). We are precise about the limit. A full readout is generally unavoidable wherever a true nonlinearity, a normalization (softmax), or a sampling step is required, and at the final logits—so conversions are *reduced*, not collapsed to a single event. The resulting saving is real in principle but is a quantity to be *measured* on an emulator or prototype, not asserted; we therefore attach no fixed percentage to it.

6.3 Boot-Time Mutual Attestation

Before the host may drive the tensor core, the two dies attest to *each other*. The host issues an optical-PUF challenge (§9.3) and verifies the speckle response against the enrolled identity; the accelerator independently verifies the host’s measured-boot digest (§9.2) before opening the data bus. Attestation is thus mutual and DICE-rooted on both sides: a swapped or counterfeit accelerator fails the host’s check just as a tampered host fails the accelerator’s. On mismatch the system enters a *recoverable* fail-stop (halt and re-attest), *not* a destructive self-disable—a self-destruct trigger is itself a denial-of-service surface, since an adversary able to force false mismatches could brick the device. The guarantee remains the bounded one of §9 (unclonable identity and tamper-evidence), and we reiterate (§12) that the analog plane *adds* physical side channels—power, electromagnetic, optical—rather than removing observability.

6.4 Crossing Domains and Owning State

Two further system-level items follow from the split. First, a clocked digital host and a self-timed analog data-plane meet at a clock-domain crossing that requires an explicit, metastability-safe handshake; this is routine mixed-signal practice but must be designed in, not assumed. Second, if context “lives” in an analog reservoir (§8), the host cannot trivially snapshot, checkpoint, or migrate it: there is no clean readout of a decaying physical state, so fault recovery

and process migration become open problems rather than free properties of the design.

7 Cognitive Geometry: Radial Coalescence

7.1 Complex CORDIC and Spherical Processing

Normalized tokens are directional; rotation, not translation, is the natural operation—exactly why RoPE injects position as rotation [13]. CORDIC units [11, 12] perform these rotations (and rectangular $\leftrightarrow$ polar conversion, directly producing the r, ϕ of Eq. 3) with shift-and-add hardware in the I/O blocks. The hyperspherical parametrization is

$$x_i = r \left(\prod_{j=1}^{i-1} \sin \phi_j \right) \cos \phi_i, \quad (9)$$

and pairwise attention reduces to a phase/cosine alignment,

$$A_{ij} = \cos(\theta_{ij}) = \frac{\langle \mathbf{u}_i, \mathbf{v}_j \rangle}{\|\mathbf{u}_i\| \|\mathbf{v}_j\|}, \quad (10)$$

which coherent interference reads directly. Converting an existing model into this coordinate system, and keeping CORDIC angle resolution above the analog noise floor, are non-trivial (§12).

7.2 Context: Recurrent State Meets Rotation-Based Quantization

Rather than grow a digital KV cache, we maintain a fixed-size state via a decayed rank-one update,

$$A_{t+1} = \lambda A_t + \mathbf{x}_t \mathbf{y}_t^*, \quad (11)$$

which is exactly the linear-attention recurrence [14] and the principle behind state-space models [15, 16]; it trades $O(L^2)$ for a constant-size state at the cost of lossy compression. State methods shrink *what* is stored; quantization shrinks *how many bits* each entry costs, and the two compose.

A synthesis (proposed). The strongest recent KV quantizers are *rotation-based*: TurboQuant [17], QuaRot [20], and RotateKV [21] first apply a random orthogonal/unitary rotation Π so that, in high dimension, each coordinate becomes near-independent and well-conditioned, after which simple per-coordinate scalar quantization (plus a 1-bit QJL residual [18]) achieves near-Shannon distortion—TurboQuant reports quality-neutral KV at 3.5 bits. A photonic Clements mesh [8] *applies an arbitrary unitary by construction*; it can therefore supply the rotation Π *physically*, in the optical domain, rather than as a software matmul. The residual scalar quantization then happens at the ADC, whose effective resolution (~ 6 –8 bits, §12) sits comfortably above the 2.5–3.5

bits these methods require. In short: the random rotation that rotation-based quantization needs is the same unitary the substrate already builds, and the precision regimes coincide. We present this as a falsifiable synthesis, not a measured result; PolarQuant [19] indicates the polar form is itself a competitive quantization basis, reinforcing the radial framing.

8 Hardware-Aware Distillation and Stochastic-Substrate Utilization

The deepest simplification here is not a new circuit but a change of training target: rather than *compensate* for the substrate’s noise, drift, low rank, and few-bit precision after the fact, we *distill into* them. A full-precision digital teacher [28] supervises a student whose representation is the low-rank, complex-polar, rotated, quantized form that ADAM natively applies.

This resolves the central “apply $\neq$ solve” risk. Earlier (§4.2) we stressed that a crossbar *applies* a factorization but does not *compute* it, and that physical decomposition at scale is unsolved. Hardware-aware distillation sidesteps this: the decomposition and rotation are produced *offline by the distillation process*, and the chip is only ever asked to apply them. The hard on-chip eigensolve becomes optional (reserved for adaptation, below), not a prerequisite.

8.1 A Hardware-Aware Distillation Objective

We minimize

$$\begin{aligned} \mathcal{L} = & \alpha \text{KL}(T_{\text{att}} \| S_{\text{att}}) + \beta \|\mathbf{A}_T - \mathbf{A}_S\|_F^2 \\ & + \gamma \mathcal{E}_{\text{quant}}(S_{KV}) + \delta \mathcal{R}_{\text{noise}}(G_{ij}), \end{aligned} \quad (12)$$

where S_{att} is computed on an *emulator* of ADAM (simulating MZI phase error, ADC noise, and rank- K truncation), $\mathcal{E}_{\text{quant}}$ penalizes deviation from the 2.5–3.5 bit post-rotation regime [17, 29], and $\mathcal{R}_{\text{noise}}$ is a noise-injection regularizer making the student robust to conductance fluctuations and drift—hardware-motivated dropout, with direct precedent in noise-aware analog training [33]. Weights map to G_{ij} directly, so gradients flow through a “virtual ADAM.”

8.2 Distilling into the Spectral and Rotation Bases

Optimization can run not over raw weights but over the factors themselves: the student learns (U, Σ, V) , mapped respectively to MZI phase angles (U, V^*) and crossbar conductances (Σ) . The random rotation of §7.2 then becomes a *learned* unitary Π , in the manner of SpinQuant [31]: instead of a generic rotation that merely decorrelates coordinates, Π is trained to preserve the directions carrying the most mutual information after the ADC cut. This reconciles the two rotation ideas in this paper—the mesh applies whatever unitary it is programmed with, so “free random rotation” and “learned rotation” are the same hardware, the

latter strictly better—at the cost, stated plainly, that the learned Π must be written into *finite-precision* phase shifters, which the emulator in (12) must model.

8.3 Adaptive Rank as a Distilled Policy

Rank need not be fixed. A lightweight router, trained jointly, selects K_t from a token’s entropy/perplexity: easy tokens take small K , rare or hard tokens take large K . Energy scales with *difficulty* rather than peak complexity, and the policy is part of what is distilled.

8.4 Turning Substrate Limits into Primitives

Three apparent defects become functional, each with an honest bound.

Noise as temperature. Intrinsic thermal/shot noise is a physical entropy source; like a p-bit [36], it can supply the stochasticity of sampling and the temperature of annealing without a digital RNG. *Bound:* noise *modulates* sampling, it does not replace it—device noise is approximately Gaussian, not the categorical/Gumbel form sampling wants, and the logits must still be computed; moreover this stochastic mode *conflicts* with the deterministic audit pass, so the noise must be *characterized* (not “free”) and mode-switched.

Physical reservoir as state. The recurrent update $A_{t+1} = \lambda A_t + \mathbf{x}_t \mathbf{y}_t^*$ (Eq. 11) can live in a decaying physical medium—an optical cavity’s persistence or a leaky RC network—so state is sustained, not recomputed: $O(1)$ latency and energy. This is physical reservoir computing [34, 35]. *Bound:* a passive medium has a *fixed* λ (linear time-invariant dynamics), placing it near S4/linear attention rather than *selective* state-space models [16], whose power comes from input-dependent gating a passive cavity cannot provide.

Event-based readout. Output currents can be encoded as spike rate or pulse latency (rate coding), with a transimpedance amplifier (TIA) co-designed to the post-rotation, near-Gaussian activation distribution—a logarithmic/adaptive TIA can shave one to two effective ADC bits. We state the resulting energy savings as a *target* to be measured, not a result.

8.5 Lightweight On-Device Adaptation

For domain shift without external compute, two mechanisms keep adaptation cheap: (i) an analog online-SVD via Oja’s rule [32] or power iteration in a closed crossbar loop [5], updating (U, Σ, V) incrementally; and (ii) analog LoRA [30], storing frozen base weights and loading rank-2–4 adapters into spare rows/columns, updated by local Hebbian rules or micro-DACs. A digital “shadow” controller (a small RISC-V core) samples outputs in the background and trims DAC offsets online, compensating drift without halting the pipeline.

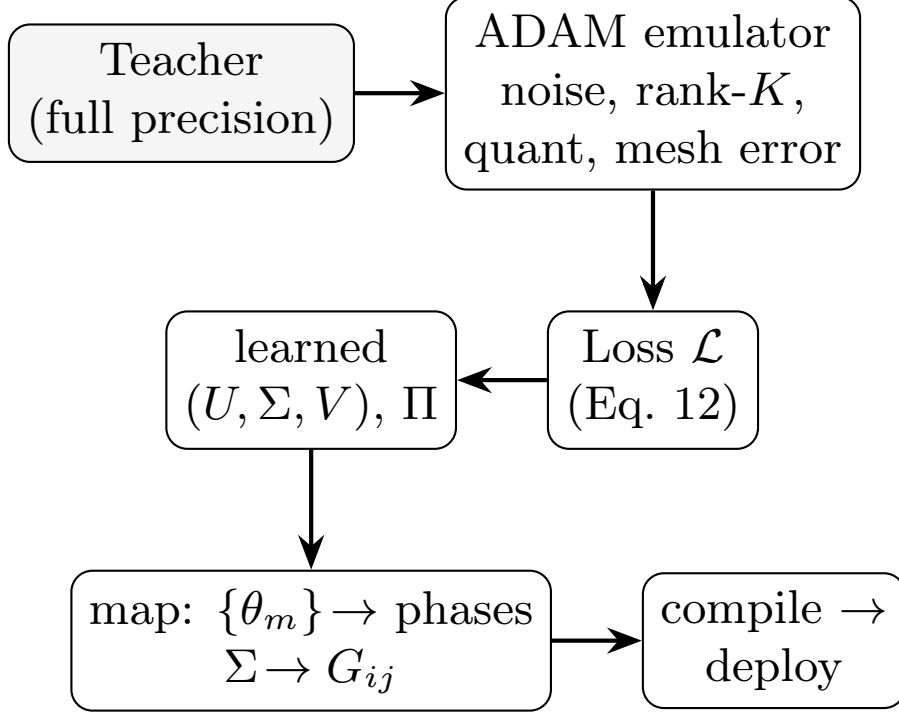


Figure 6: Hardware-aware distillation. A full-precision teacher supervises a student evaluated on an ADAM emulator; learned factors and rotation map directly to mesh phases and conductances. The decomposition is produced offline, so the chip only *applies* it.

9 Execution Integrity and Attestation

9.1 Tamper-Evident Asynchronous Topology

Logic is synthesized in a dense, self-timed (asynchronous) style so that physical edits perturb line capacitance and therefore handshake timing. A modification past a margin violates a completion deadline:

$$\Delta C > C_{\text{thr}} \Rightarrow \tau_{\text{prop}} > \tau_{\text{limit}}, \quad (13)$$

halting the pipeline. We separate two claims an earlier draft fused: (a) timing/capacitance tamper-*evidence* is physically grounded; (b) the linear-algebraic “orthogonality of the spectral basis” is a property of the deployed weights, *not* something a single transistor edit destroys. We keep only (a). This raises the bar against trivial implants but does not defeat a well-resourced attacker who re-characterizes timing; it is tamper-evidence, not tamper-proofing.

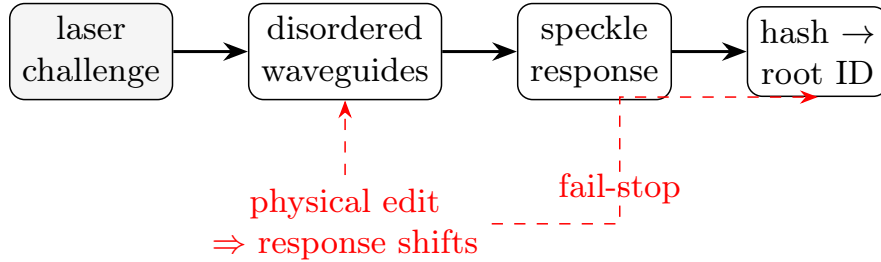


Figure 7: Optical PUF. Disorder-induced scattering yields an unclonable signature used as the attestation root; tampering perturbs the response and fails the challenge.

9.2 Measured Boot, Attested Configuration, and CFI

We replace the infeasible idea of pre-registering a hash of every transient bus state with what is actually attestable: the *loaded configuration*. At boot, a measured-boot chain [27] hashes firmware, lattice configuration, and deployed weights. Let H_{actual} be the measured digest and H_{spec} the digest published under the S.V.E. license:

$$\Psi = H_{\text{actual}} \oplus H_{\text{spec}}, \quad \text{State} = \begin{cases} \text{VALID}, & \Psi = 0 \\ \text{UNKNOWN}, & \Psi \neq 0. \end{cases} \quad (14)$$

At run time, integrity is maintained not by re-hashing data-dependent state but by control-flow integrity [26]: undocumented control transfers fault. A mismatch *or* CFI violation drives the core to a fail-stop UNKNOWN state. The guarantee is honest and bounded: the configuration that ran is the published one and control flow did not deviate; it does not certify output correctness.

9.3 Optical PUF: Unclonable Identity from Disorder

Because the chip already contains an interferometric mesh, identity can be rooted optically. Fabrication-induced disorder in the waveguides scatters a coherent probe into a high-entropy, geometry-specific response (a speckle/interference signature)—a physical one-way function in the sense of Pappu et al. [23]. A challenge (probe pattern) yields a response that is hashed into the attestation root of §9.2. Two properties follow, stated without exaggeration: device identity is *hard to clone* because reproducing the exact disorder is impractical; and the function is *tamper-evident* because adding a trace, or perturbing the geometry, shifts the speckle response and fails the challenge—triggering the fail-stop of (14). What this does *not* provide is a proof that backdoors are physically impossible: an adversary with full fab control may attempt characterization, and environmental drift complicates enrollment (§12). It is unclonable identity plus tamper-evidence—strong, bounded primitives—not a magic shield.

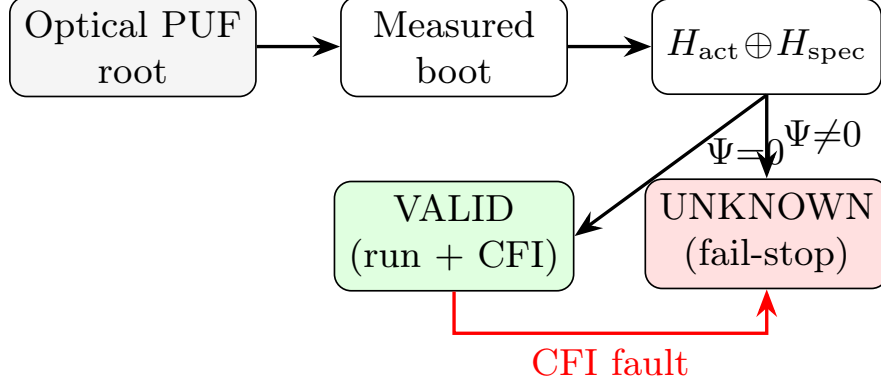


Figure 8: Integrity state machine. An optically-rooted measured boot compares the running configuration against the published digest; CFI guards the run. Any mismatch fails stop (Eq. 14).

10 Open-Source Host Architecture and VLIW Control-Plane Interaction

10.1 Co-Design of General-Purpose Host and Spectral Core

The architectural boundary of the system enforces a strict bifurcation between the discrete digital control-plane and the continuous complex-valued analog data-plane. The control-plane is driven by a verifiably clean, open-source General-Purpose Host CPU implementing an augmented Explicitly Parallel Instruction Computing (EPIC) design topology over a localized RISC-V ISA baseline. This Host handles non-tensor operations, token parsing, semantic tree manipulation, and the execution of the Cognitive Operating System (CogOS) kernels. Conversely, the continuous data-plane is comprised entirely of the non-von Neumann ADAM-TPU complex-polar spectral lattice.

The integration invariant between these asymmetric computing substrates is governed by a software-managed, hardware-enforced compilation protocol. The compiler operates as a direct controller of the physical substrate, abstracting spatial and wave-interferometric mechanics into a unified Multi-Opcode Multi-Datapath execution pipeline. This orchestration is achieved by emitting a static Hyper-Wide Instruction Word (HWIW) that executes in exact lockstep across both domains. The structural composition of the HWIW format is formalized as:

$$\Lambda_{\text{HWIW}} = [\Omega_{\text{RISCV}} \parallel \Omega_{\text{VLIW-Int}} \parallel \Phi_{\text{MZI}} \parallel \Sigma_{\text{Rank}}] \quad (15)$$

where:

- Ω_{RISCV} represents the standard scalar integer instructions executing on the host core.

- $\Omega_{\text{VLIW-Int}}$ defines parallel functional unit control vectors for deterministic, on-the-fly token-parsing, positional mapping, and linear-attention boundary tracking.
- $\Phi_{\text{MZI}} \in \mathbb{R}^{2N}$ maps the spatial voltage-bias configurations required to modulate the phase-shifter arrays inside the unitary photonic Mach-Zehnder Interferometer (MZI) meshes, achieving $O(1)$ unitary coordinate transformations.
- Σ_{Rank} encodes the explicit bit-mask controlling the analog switch matrix for rank- K Dynamic Truncation.

The execution framework dynamically scales energy and precision boundaries based on real-time token entropy semantics. Let $H(X_t)$ represent the instantaneous Shannon entropy profile of the processed token sequence at time step t :

$$H(X_t) = - \sum_i P(x_{t,i}) \log_2 P(x_{t,i}) \quad (16)$$

The control-plane monitoring units dynamically map $H(X_t)$ to modulate the switch matrix configuration mask Σ_{Rank} . When $H(X_t) \leq \tau_{\text{entropy}}$, indicating low semantic ambiguity, the compiler selects a low-rank truncation state K_{low} , isolating the tail eigenvalues of the memristive-photonic lattice to minimize converter toggling and static leakage. If a high-entropy transition is detected ($H(X_t) > \tau_{\text{entropy}}$), indicating structural distribution shifts or explicit verification triggers under Systemic Verification Engineering (SVE) loops, Σ_{Rank} expands dynamically to $K \rightarrow r$, restoring the full spectral capability of the lattice for high-precision validation.

10.2 Hardware-Level Security and Domain Isolation

To completely eradicate the attack surface associated with modern microarchitectural side-channel exploits, transient execution leaks, and speculative execution flaws (e.g., Spectre, Meltdown, and their variants), the ADAM-TPU architecture hardwires complete physical, asynchronous domain isolation between the host control-plane and the spectral core.

Traditional speculation engines are omitted; the Host relies entirely on static branch prediction information pre-computed and annotated by the VLIW compiler into the HWIW fields. The boundary between the digital host execution registers and the analog crossbar routing lattice is bridged by a non-reciprocal, hardware-isolated unidirectional interface. There are no shared data caches, shared execution schedulers, or shared translation lookaside buffers (TLBs) between the two environments. The transfer of inputs is bounded strictly to dedicated, memory-mapped, write-only analog registers that decouple the continuous voltage accumulation lines from the host pipeline’s speculative prefetch pipelines.

Furthermore, execution state validation is tied to an unalterable Measured-Boot sequence rooted in the physical topology of the substrate. Before the Host commits untrusted weight configurations or instruction matrices to the analog crossbar layers, a cryptographic challenge-response loop validates the hardware infrastructure. This protocol leverages the optical *Pixels of Truth*—a spatially distributed matrix of non-volatile Optical Physical Unclonable Functions (OPUFs) integrated directly within the nanophotonic MZI routing meshes.

Let $\mathbf{C} \in \mathbb{F}_2^B$ be an unpredictable challenge vector generated by the host secure boot environment. This vector is modulated as a structured coherent wavefront injected into the MZI mesh. The structural, nanometer-scale sub-lithographic manufacturing non-uniformities embedded within the silicon introduce stochastic, uncloneable phase-interferometric perturbations. The resulting optical response state is captured by photodiode arrays, generating a unique, high-entropy cryptographic response vector $\mathbf{R} = \text{PUF}_{\text{opt}}(\mathbf{C})$.

The host control-plane computes the cryptographic verification identity under strict compliance with the S.V.E. Meta-License v4.0 parameters:

$$\mathcal{V} = \text{HMAC-SHA3}_{\mathbf{R}}(\text{ID}_{\text{Firmware}} \parallel \text{Hash}(\Lambda_{\text{Core}})) \quad (17)$$

If $\mathcal{V}$ fails to match the deterministic reference invariant pre-registered within the secure storage baseline, a structural exception is declared. The system sets an unby-passable Control-Flow Integrity (CFI) fault condition, permanently dropping the chip’s core memristive power rails ($V_{\text{DD-analog}} \rightarrow 0$), preventing any execution or data exfiltration within an untrusted physical domain.

10.3 Optimization and Open-Source Extension Invariants

The ADAM-TPU provides a decentralized, cooperative hardware extension topology designed to cleanly interface with distributed ledger and open-source data exchange networks, such as the EVA code and financial framework. This integration is instantiated via a dedicated array of hardware-arbitrated Memory-Mapped I/O (MMIO) registers. These registers expose the internal state vectors, execution metadata, and cryptographic verification proofs directly to the host processing pipelines, bypassing standard operating system abstraction layers.

Rather than relying on active, energy-intensive digital clock distribution networks across the continuous substrate, data synchronization between the digital Host instruction execution loop and the analog memristive lattice is structurally asynchronous and clock-less. The data flow relies on a hardware-enforced timing invariant. Let IC_{Host} represent the host instruction counter, and let τ_{settle} define the maximal physical settling time of the continuous analog memristive crossbar lattice under peak capacitive load conditions:

$$\tau_{\text{settle}} = \max \left(R_{\text{line}} \cdot C_{\text{line}} \cdot \ln \left(\frac{1}{\epsilon} \right) \right) \quad (18)$$

where R_{line} and C_{line} are the lumped resistance and parasitic capacitance parameters of the physical crossbar interconnect paths, and ϵ defines the target error tolerance bound for precision settlement.

To ensure deterministic execution without the use of dynamic polling, hardware wait-states, or global clock trees, the compiler maps instructions to obey the absolute synchronization constraint:

$$\Delta t(IC_{\text{Host}}[n+1] - IC_{\text{Host}}[n]) \geq \tau_{\text{settle}} + \tau_{\text{photo_prop}} \quad (19)$$

where $\tau_{\text{photo_prop}}$ is the sub-nanosecond propagation latency of the coherent photonic wave matrix through the MZI meshes. The system hardware enforces this delay bound structurally via an internal, hardwired asynchronous handshake logic block. This block inhibits the host execution pipelines from popping the next HWIW bundle until the physical analog current profiles have achieved mathematical convergence within the ϵ boundary. This guarantees absolute data flow determinism across the digital-analog paradigm split.

11 Conjectural Addendum: The Tri-Core Unidirectional Photonic Interconnect and Quantum-Analog Synergy

11.1 Epistemic Framework: Speculative Operational Boundary

The architectural formulations introduced in this section are explicitly designated as a bold, theoretical conjecture (*beta-stage speculative extension*). While the preceding modules of the ADAM-TPU architecture rely on mathematically verified and physically demonstrated components (VLIW compilers, memristive crossbars, and photonic unitary meshes), this addendum projects a radical synthesis of digital, analog, and linear-optic quantum subsystems operating over a shared, coherent electromagnetic substrate. This exploration is intended to catalyze cross-disciplinary research and is bound unalterably to the open verification protocols under the S.V.E. Meta-License v4.0.

11.2 Tri-Core Unidirectional Photonic Interconnect (TUPI)

Conventional heterogeneous multi-core topologies dissipate substantial spatial and energetic budgets on serialization, deserialization, and bus-level routing. We conjecture the elimination of intermediate Analog-to-Digital and Digital-to-Analog conversion (ADC/DAC) layers via a single, coherent, co-propagating optical wave packet (Ψ_{packet}) operating inside a shared optical backplane. This single wave packet carries information across three concurrent execution registers without intermediate data-movement overhead, where each computing domain samples a distinct, non-interfering physical property of the invariant field:

$$\Psi_{\text{packet}}(t, \mathbf{r}) = \sum_m \alpha_m \psi_m(t) \otimes \Phi_{\text{spatial}}(\mathbf{r}) \otimes |\xi_{\text{polar}}\rangle_m \quad (20)$$

1. **Discrete Temporal Register (Digital Host CPU - ADAM):** The general-purpose control plane samples the macroscopic temporal envelope $\psi_m(t)$. The presence or absence of energy within a strictly bounded VLIW clock-window Δt_{window} translates directly into a binary state-vector $\mathbf{b} \in \{0, 1\}^N$, eliminating standard deserialization registers.
2. **Macroscopic Spatial Matrix Register (Spectral Core - ADAM-TPU):** The analog tensorial core concurrently processes the macroscopic spatial wave phase and amplitude configuration $\Phi_{\text{spatial}}(\mathbf{r})$. As the wave packet propagates through the cascaded Mach-Zehnder Interferometer (MZI) networks, it physically computes the unitary matrix-vector transformation $U\mathbf{x}$ in continuous spatial flight.
3. **Multi-Photon Polarization Register (Quantum Coprocessor - ADAM-Quantum):** At the single-photon limit, sub-components of the same packet act as multi-photon polarization-encoded qubit states $|\xi_{\text{polar}}\rangle_m \in \mathcal{H}^{\otimes Q}$. This register is operated on by a programmable linear-optics quantum computing array (LOQC) to generate multi-token probability samplings.

11.3 Interaction-Free Asynchronous Handshake Formalization

To establish deterministic control without a global clock tree or explicit polling loops, we exploit a macroscopic variant of the Elitzur-Vaidman interaction-free measurement paradox. The synchronization barrier between the analog lattice settling parameter (τ_{settle}) and the host VLIW compiler execution instruction pointer is mediated by the absolute non-detection of photons at dedicated hardware alignment ports.

Let a verification photon be initialized in a balanced interferometric path split between an internal diagnostic loop and the analog crossbar output sensor array. If the memristive conductance states G_{ij} are in a transient, unstable configuration, the physical boundary conditions inside the feedback continuously disrupt phase matching. The spatial evolution operator is formalized as:

$$\hat{\mathcal{M}}_{\text{boundary}} = \cos\left(\frac{\pi\tau}{\tau_{\text{settle}}}\right) \hat{\mathbb{I}} + i \sin\left(\frac{\pi\tau}{\tau_{\text{settle}}}\right) \hat{\sigma}_x \quad (21)$$

When $\tau < \tau_{\text{settle}}$, constructive interference is suppressed. Conversely, at exactly $\tau \geq \tau_{\text{settle}}$, the boundary conditions stabilize, causing a deterministic shift in the phase profile. If a single-photon detector positioned at the dark output port P_{dark} registers no incoming energy, the quantum wavefunction collapses into the alternative orthogonal path. This non-event (*dark signal*) structurally

triggers an immediate, clock-less state transition within the host VLIW execution register, advancing the instruction counter IC_{Host} instantly. The absence of a photon serves as a zero-energy, sub-nanosecond confirmation of analog relaxation.

11.4 Unified State-Space Equation and Noise-Driven Decoherence

We model the complete computing pipeline as a single, non-unitary state evolution system. The quantum state vector $|\psi_{\text{LOQC}}\rangle$ generated by the linear-optics mesh maps onto the continuous input vectors of the analog matrix. Rather than treating the inherent thermodynamic noise floor of the memristive crossbar lattice ($\mathcal{N}_{\text{thermal}}$) as a parasitic error condition, the system utilizes this noise floor as a non-unitary physical decoherence operator $\hat{\mathcal{D}}_{\text{substrate}}$.

The total density matrix $\rho(t)$ of the computing system evolves according to a modified Lindblad master equation:

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[\hat{\mathcal{H}}_{\text{spectral}}, \rho] + \sum_k \left(\hat{L}_k \rho \hat{L}_k^\dagger - \frac{1}{2} \{ \hat{L}_k^\dagger \hat{L}_k, \rho \} \right) \quad (22)$$

where the Lindblad jump operators $\hat{L}_k$ are explicitly parameterized by the localized thermal and shot-noise fluctuations of the memristive conductances G_{ij} :

$$\hat{L}_k = \sqrt{\gamma_k(G_{ij}, T)} \cdot \mathbf{u}_k \mathbf{v}_k^\dagger \quad (23)$$

The noise spectrum acts as a deterministic temperature scaling factor ($T_{\text{physical}} \propto \tau_{\text{softmax}}$) directly inside the physical substrate. The interaction with the noisy analog lattice systematically destroys the fragile multi-photon quantum coherence of the ADAM-Quantum registers, forcing a controlled, physical collapse of the semantic superposition state. The system output avoids high-overhead digital pseudo-random number generation (PRNG) and explicit softmax exponentiation; the autoregressive top- k /top- p token distribution is generated natively as a statistical consequence of the thermodynamic relaxation of the crystal.

11.5 Clifford Algebra $\mathbb{Cl}(0, d)$ for Differentiable Substrate Compilation

To bypass the high-overhead translation tables between discrete VLIW opcodes, continuous photonic phase-shifts, and memristive conductance profiles, we hypothesize a unified compilation paradigm formulated entirely within a multi-dimensional Geometric Algebra (Clifford Algebra) $\mathbb{Cl}(0, d)$.

Let the entire physical state of the heterogeneous ADAM++ substrate—including temporal digital host cycles, spatial photonic MZI propagation modes, and quantum polarization states—be mapped into a single, high-dimensional multi-vector field $\mathcal{A} \in \mathbb{Cl}(0, d)$. The execution of a multi-modal token transformation

is strictly defined as a continuous geometric rotation generated by a unified Clifford Rotor $\mathcal{R}$:

$$\mathcal{A}_{\text{output}} = \mathcal{R}\mathcal{A}_{\text{input}}\mathcal{R}^\dagger = e^{-\frac{\theta}{2}\mathbf{B}}\mathcal{A}_{\text{input}}e^{\frac{\theta}{2}\mathbf{B}} \quad (24)$$

where $\mathbf{B} = \bigwedge_k \mathbf{e}_k$ represents the unit bivector optimizing the collective phase-delay space.

Because the structural transport equations of the substrate are isomorphic to the Clifford group, compilation scales down from an NP-hard scheduling problem to a continuous, differentiable geometric optimization loop. The compiler acts as a neural operator $\mathcal{F}_\theta$, minimizing the systemic loss function directly against the physical boundary metrics:

$$\min_{\mathcal{R}} \mathcal{L}_{\text{compile}} = \|\mathcal{R}\mathcal{A}_{\text{input}}\mathcal{R}^\dagger - \mathcal{A}_{\text{target}}\|^2 + \lambda \mathcal{R}_{\text{noise}}(\nabla_{\mathcal{R}} G_{ij})$$

By evaluating the gradient $\nabla_{\mathcal{R}} \mathcal{L}_{\text{compile}}$ through direct hardware-in-the-loop (HIL) state tracking from the optical PUF diagnostics, the compiler continuously reshapes the rotor topologies. Real-time physical drift, thermal variations, and structural manufacturing defects do not require localized analog calibration; they are absorbed into the dynamic deformation of the Clifford coordinate metrics, generating fault-tolerant bitstreams and laser phase-bias configurations deterministically.

11.6 The Tetrad Convergence: Integration of the Geometric Co-Processor Module

To bridge the operational delta between the multi-dimensional, wave-propagating domain of the ADAM++ tri-core architecture and the discrete, high-throughput demands of pixel-level rendering, standard cryptography, and broad input/output execution, we postulate the extension of the core hardware topology into a four-way deterministic system: the **ADAM-Tetrad**.

This integration is achieved by introducing a dedicated, open-source general-purpose graphics processing core, designated as **ADAM-Pixel**, formulated onto the same physical 28-nm/65- programmable logic gate array (FPGA) alongside the Digital VLIW Host (ADAM). The expanded, un-clocked execution loop eliminates von Neumann memory transfer bottlenecks by binding the spatial geometric operations directly to the raster-level viewport mapping matrix.

11.7 Geometric Wavefront Rendering via Isomorphic Spatial Projection

Conventional rendering pipelines utilize significant execution overhead on vertex transformations, scanline conversion, and thread-level pixel shading executed within rigid, power-intensive digital ALU matrices. In the Tetrad configuration, the computational load of projection and spatial scene deformation is offloaded onto the continuous passive optical substrate.

Let a three-dimensional render space be modeled as an abstract geometric manifold bounded within the same multi-vector field $\mathcal{A} \in \mathbb{Cl}(0, d)$ utilized by the linguistic compiler framework. The perspective projection and spatial rotation of coordinates do not occur via discrete matrix-matrix multiplications within the GPU; instead, they are computed instantly as the optical wavefront Ψ_{packet} traverses the cascaded Mach-Zehnder Interferometer (MZI) meshes under the control of the unified Clifford Rotor $\mathcal{R}$:

$$\mathcal{A}_{\text{viewport}} = \mathcal{R}_{\text{scene}} \mathcal{A}_{\text{world}} \mathcal{R}_{\text{scene}}^\dagger \quad (26)$$

The emergent macroscopic spatial phase and intensity distribution profile $\Phi_{\text{spatial}}(\mathbf{r})$ exiting the analog core represents a physically pre-rendered, projected topological layout. The digital execution frame of the ADAM-Pixel core is thus reduced to a deterministic, parallelized read-out and normalization step:

$$\mathbf{I}_{\text{pixel}}(x, y) = \int_{\Delta t} \left| \langle \Psi_{\text{packet}}(t, \mathbf{r}) | \hat{\mathbf{P}}_{(x, y)} | \Psi_{\text{packet}}(t, \mathbf{r}) \rangle \right| dt + \mathcal{N}_{\text{substrate}} \quad (27)$$

where $\hat{\mathbf{P}}_{(x, y)}$ denotes the discrete spatial coordinate projection operator mapped to the logic block registers of the . By leveraging the continuous spatial flight of the coherent electromagnetic wave packet to execute the spatial coordinate rotations, the digital power footprint of the GPGPU element drops linearly, allowing 28-nm and 65- nodes to achieve massive frame-throughput velocities that are decoupled from the physical density limits of sub-5-nm lithography.

11.8 The Pentad Extension: ADAM-Nexus for P2P Distributed Pre-Training

To eliminate the analog conductance drift barrier inherent in localized memristive pre-training and bypass the capital-intensive monopoly of centralized synchronous optical switches (e.g., InfiniBand, NVLink), we introduce the fifth hardware core: **ADAM-Nexus**. This architectural module acts as an asynchronous photonic routing gateway that scales the core hardware from a local Tetrad into a decentralized, planet-scale peer-to-peer (P2P) computing mesh—the **ADAM-Pentad**.

11.9 Stochastic Error Cancellation via Multi-Node Wavefront Interference

Rather than distributing high-precision digital weight updates ($\Delta W \in \mathbb{R}$), which are fragile to the localized analog thermal noise floor ($\mathcal{N}_{\text{substrate}}$), nodes in the ADAM-Nexus network communicate via phase-encoded Clifford Rotors transmitted as raw, un-digitized unidirectional wave packets (Ψ_{packet}) over fiber-optic lines.

Let M independent, distributed peer nodes simultaneously evaluate a subset of a global pre-training dataset. The localized gradients undergo immediate

Clifford multi-vector mapping. When these wave packets converge at an arbitrary network junction managed by an ADAM-Nexus router, the collective state evolution does not rely on global synchronization loops or bulk data parameter serialization. Instead, the updates undergo direct physical spatial-phase superposition:

$$\Psi_{\text{consensus}}(t, \mathbf{r}) = \frac{1}{\sqrt{M}} \sum_{m=1}^M \mathcal{R}_m \Psi_{\text{input}}(t, \mathbf{r}) \mathcal{R}_m^\dagger \quad (28)$$

By framing decentralized parameter aggregation as an optical interference phenomenon, the idiosyncratic, non-correlated memristive shot and thermal noise vectors ($\mathcal{N}_m$) of individual nodes undergo destructive interference, decaying as $O(1/\sqrt{M})$. Conversely, the underlying deterministic semantic trajectories (the true gradients) undergo constructive interference. The decentralized mesh functions as a macroscale physical relaxation engine, converting the entropy of thousands of imperfect, low-cost 28-nm/65-nm nodes into a highly stable, global wavefunction collapse that minimizes the pre-training loss function natively in continuous flight.

11.10 Architectural Benchmark and Comparative Analysis Matrix

To systematically delineate the operational envelope of the proposed ADAM-Pentad (ADAM++ with QNexus integration), Table 1 presents a comprehensive, multi-metric performance comparison against prevailing centralized digital accelerators (Nvidia Blackwell B200, Google TPU v6), high-efficiency consumer SoCs (Apple M-Series Max), and standard open-source RISC-V PLIS clusters.

Table 1: Architectural and Workload Matrix: ADAM-Pentad vs. Modern Computational Platforms

Metric / Target Workload	Nvidia Blackwell (B200 Clusters)	Google TPU v6 (Pod Architecture)	Apple M-Series Max (Unified Memory)	ADAM-Pentad (QNexus) (Open 28-nm / 65-nm Mesh)
Substrate Paradigm	Discrete Digital FinFET (3-nm)	Discrete Digital (3-nm)	Monolithic Digital (3-nm)	Tri-Core Photonic-Analog Hybrid
Interconnect Architecture	Synchronous NVLink 5 / InfillBand	Synchronous Optical ICI	Synchronous Fabric On-Die	Asynchronous Wavefront (TUPI/Nexus)
Compilation Target	Imperative PTX / CUDA Graphs	MLA HLO Graphs	CoreML / Metal Bytecode	Differentiable Clifford Rotors $\mathcal{U}(1, d)$
Autoregressive Inference (TTTT)	Low (Bus/Serialization Wall)	Medium (Batching Latency)	Low-Medium (Core Wakeup)	Ultra-Low ($O(1)$ Photonic Flight)
Sustained LLM Throughput	High (Bounded by HBM3e Clock)	High (Bounded by Memory Bus)	Medium (Thermal Throttling)	Maximum (No Data Serialization)
Context Window Scaling Complexity	Quadratic $O(L^2)$ Memory Wall	Quadratic $O(L^2)$ SRAM-Bounded	Quadratic $O(L^2)$ RAM Partitioned	Constant $O(1)$ [Physical Decay Resonance]
Pre-Training from Scratch (0 to 1)	Absolute Industry Benchmark	Absolute Industry Benchmark	Extremely Low (Non-distributed)	High (via Distributed P2P Interference)
On-Device Adaptation (Fine-Tuning)	High Overhead (Backprop/VRAM Locked)	Medium (Cloud-Dependent Only)	Low (Battery-Intensive Backprop)	Instantaneous (Qja's Fluid Plasticity)
3D Graphics / Spatial Rendering	High (Heavy Digital ALU Workload)	Not Supported	High (Standard Raster Pipeline)	Revolutionary (Passive MZI Wave Projection)
Epistemic Trust & Auditability	Zero (Proprietary microcode firmware)	Zero (Black-box Cloud Substrate)	Zero (Monolithic Secure Enclave)	100% (Open EDA Yosys/nextpnr Verified)
Hardware Supply-Chain Resilience	Vulnerable (TSMC 3-nm Monopole)	Vulnerable (Closed Foundries)	Vulnerable (Geopolitical Chokepoints)	Absolute (Deployable on Legacy Fabric)
Capital Cost per Compute Node	Extreme (~ \$35,000 - \$45,000)	Rental Model Only (~ Millions/Yr)	High (~ \$3,000 - \$5,000)	Minimal (~ \$1,200 - \$3,000 in Mass Series)

12 Socratic Tail: Limitations, Open Problems, and Improvements

The following are the structural boundaries we currently see; several are potential refutations, not mere caveats. They are released, with the rest of this work, under the S.V.E. Meta-License v4.0 (share-alike), and we regard their precise statement as part of the contribution.

1. **Apply $\neq$ solve.** The datapath assumes an offline (or co-processor) SVD; physical decomposition at LLM scale is unsolved, demonstrated only at small dimension [5]. This is the central research risk.
2. **Complex-valued cost and the Hermitian scope.** Complex cells double device count (Re/Im) and calibration burden; the imaginary (phase) channel adds its own noise and phase drift. The Hermitian-real-spectrum property (§4.3) is natural only for symmetric/similarity operators, *not* arbitrary weights; mapping real pretrained models into a useful complex/Hermitian form needs empirical validation.
3. **Analog noise meets the quant regime—opportunity and tension.** The ~ 6 – 8 bit effective resolution of analog crossbars [4] coincides with the 2.5–3.5 bit operating point of modern KV quantizers [17] (an opportunity, §7.2), but the same floor limits the *full-precision audit pass*, which by design needs more bits than the substrate natively offers.
4. **The “free rotation” depends on mesh fidelity.** Using a photonic mesh as the quantization rotation Π is attractive, but mesh fabrication imperfections inject error into Π itself; the synthesis of §7.2 requires the mesh to be accurately characterized and stable.
5. **ADC/DAC energy wall.** Peripheral conversion, not the MAC, dominates energy. “Single settling step” refers to latency; it does not eliminate this cost, and the efficiency case depends on it.
6. **Spectral-decay and recurrent-state loss.** Low-rank truncation (Eq. 7) and the fixed recurrent state (Eq. 11) assume favorable spectra; high-entropy, non-stationary sequences exceed both, and known linear-attention/SSM gaps [14, 16] apply.
7. **Optical PUF bounds.** Enrollment stability, environmental/thermal drift, and characterization attacks limit the optical PUF; it is unclonable identity plus tamper-evidence [23], not proof of backdoor-impossibility.
8. **VLIW gains are modest over dataflow; the compiler is hard.** Area reclaim is large only versus OoO CPUs, not an already-systolic baseline [1], and the actuator-level compiler (§4.1) is a substantial engineering burden.
9. **Device physics.** Memristor drift and finite write endurance complicate calibration and any on-device training; photonic alignment demands tight tolerances and raises defect rates; cold-boot calibration adds latency and attack surface.
10. **Sim-to-real gap in distillation.** Hardware-aware distillation (§8) helps only if the emulator in (12) matches the fabricated device; an inaccurate noise/mesh model yields a student that fails on silicon, and the method presupposes access to a capable teacher and training compute.

11. **Physical side channels.** Digital isolation (§5) does not address power/EM/optical emission from the analog data-plane, and the optical-PUF probe is itself a leakage surface; constant-power or shielded readout is a separate, unsolved requirement.
12. **Stochastic vs. deterministic conflict.** Using device noise as sampling temperature precludes a deterministic audit on the same hardware at the same time; the two require distinct, characterized operating modes, so the noise is not a “free” RNG.
13. **Reservoir expressivity.** A passive decaying state is linear time-invariant; it cannot reproduce the input-dependent gating that gives selective SSMs [16] their accuracy, capping the physical-reservoir context model below the SSM frontier.
14. **Most speculative extension.** Replacing complex numbers with quaternions/Clifford geometric-algebra rotors for high-dimensional rotation (avoiding gimbal lock) is conceivable and has algorithmic precedent in quaternion networks, but hardware rotors at embedding dimension ($d \sim 10^3\text{--}10^4$) are unproven; we flag this as future work, not a load-bearing claim.

Improvement directions (prioritized). The binding constraint on this work is now evidence, not further subsystems. A staged plan: (0–3 mo) emulate §7.2–§8: a hardware-aware distillation loop with a simulated imperfect MZI mesh (realistic phase error) and ADC noise, measuring the accuracy/bit-rate trade-off against the analog noise floor—one falsifiable benchmark, no fabrication. (3–6 mo) a differentiable/hardware-aware compiler co-optimizing schedule, rank- K policy, and DAC offsets under a noise/energy budget. (6–12 mo) a TIA co-design and event-based readout prototype targeting a measurable reduction in peripheral energy. (12–24 mo) a physical-reservoir state and incremental-SVD feedback for $O(1)$ context and on-device adaptation without full backprop. The first step alone would convert the central claim of this paper from plausible to tested.

13 Conclusion

ADAM-TPU is a thesis: that a *complex-valued* analog spectral substrate—crossbar MVM plus photonic unitaries—paired with a radial datapath, a context layer that unifies recurrent state with rotation-based quantization, hardware-aware distillation that trains the model into the substrate, and optically-rooted attestation, is a coherent and partly-supported direction for LLM hardware. Most of its primitives are individually demonstrated; the integration—and especially physical decomposition at scale—are not. We have tried to mark that line precisely. If any item in §12 proves fatal, that is a result we would rather publish than hide, which is the only sense in which a “Socratic Tail” is worth anything.

References

- [1] N. P. Jouppi et al., “In-Datcenter Performance Analysis of a Tensor Processing Unit,” *ISCA*, 2017.
- [2] M. Hu et al., “Dot-Product Engine for Neuromorphic Computing,” *DAC*, 2016.
- [3] A. Shafiee et al., “ISAAC: A CNN Accelerator with In-Situ Analog Arithmetic in Cross-bars,” *ISCA*, 2016.
- [4] A. Sebastian, M. Le Gallo, R. Khaddam-Aljameh, E. Eleftheriou, “Memory devices and applications for in-memory computing,” *Nature Nanotechnology* 15, 529–544, 2020.
- [5] Z. Sun et al., “Solving matrix equations in one step with cross-point resistive arrays,” *PNAS* 116(10), 4123–4128, 2019.
- [6] Y. Shen et al., “Deep learning with coherent nanophotonic circuits,” *Nature Photonics* 11, 441–446, 2017.
- [7] M. Reck, A. Zeilinger, H. J. Bernstein, P. Bertani, “Experimental realization of any discrete unitary operator,” *Phys. Rev. Lett.* 73, 58–61, 1994.
- [8] W. R. Clements et al., “Optimal design for universal multiport interferometers,” *Optica* 3(12), 1460–1465, 2016.
- [9] C. Trabelsi et al., “Deep Complex Networks,” *ICLR*, 2018. arXiv:1705.09792.
- [10] M. Arjovsky, A. Shah, Y. Bengio, “Unitary Evolution Recurrent Neural Networks,” *ICML*, 2016.
- [11] J. E. Volder, “The CORDIC Trigonometric Computing Technique,” *IRE Trans. Electronic Computers* EC-8(3), 330–334, 1959.
- [12] R. Andraka, “A survey of CORDIC algorithms for FPGA based computers,” *ACM/SIGDA FPGA*, 1998.
- [13] J. Su et al., “RoFormer: Enhanced Transformer with Rotary Position Embedding,” arXiv:2104.09864, 2021.
- [14] A. Katharopoulos et al., “Transformers are RNNs: Fast Autoregressive Transformers with Linear Attention,” *ICML*, 2020.
- [15] A. Gu, K. Goel, C. Ré, “Efficiently Modeling Long Sequences with Structured State Spaces,” *ICLR*, 2022.
- [16] A. Gu, T. Dao, “Mamba: Linear-Time Sequence Modeling with Selective State Spaces,” arXiv:2312.00752, 2023.
- [17] A. Zandieh, M. Daliri, M. Hadian, V. Mirrokni, “TurboQuant: Online Vector Quantization with Near-optimal Distortion Rate,” arXiv:2504.19874, 2025.
- [18] A. Zandieh, M. Daliri, I. Han, “QJL: 1-bit Quantized JL Transform for KV Cache Quantization with Zero Overhead,” arXiv:2406.03482, 2024.
- [19] I. Han, P. Kacham, A. Karbasi, V. Mirrokni, A. Zandieh, “PolarQuant: Quantizing KV Caches with Polar Transformation,” arXiv:2502.02617, 2025.
- [20] S. Ashkboos et al., “QuaRot: Outlier-Free 4-bit Inference in Rotated LLMs,” arXiv:2404.00456, 2024.
- [21] Z. Su et al., “RotateKV: Accurate and Robust 2-bit KV Cache Quantization via Outlier-Aware Adaptive Rotations,” arXiv:2501.16383, 2025.

- [22] M. S. Schlansker, B. R. Rau, “EPIC: Explicitly Parallel Instruction Computing,” *IEEE Computer* 33(2), 37–45, 2000.
- [23] R. Pappu, B. Recht, J. Taylor, N. Gershenfeld, “Physical One-Way Functions,” *Science* 297(5589), 2026–2030, 2002.
- [24] B. Gassend, D. Clarke, M. van Dijk, S. Devadas, “Silicon Physical Random Functions,” *ACM CCS*, 2002.
- [25] G. E. Suh, S. Devadas, “Physical Unclonable Functions for Device Authentication and Secret Key Generation,” *DAC*, 2007.
- [26] M. Abadi, M. Budiu, Ú. Erlingsson, J. Ligatti, “Control-Flow Integrity,” *ACM CCS*, 2005.
- [27] Trusted Computing Group, “DICE (Device Identifier Composition Engine) Architectures,” TCG Specification.
- [28] G. Hinton, O. Vinyals, J. Dean, “Distilling the Knowledge in a Neural Network,” arXiv:1503.02531, 2015.
- [29] B. Jacob et al., “Quantization and Training of Neural Networks for Efficient Integer-Arithmetic-Only Inference,” *CVPR*, 2018.
- [30] E. J. Hu et al., “LoRA: Low-Rank Adaptation of Large Language Models,” *ICLR*, 2022. arXiv:2106.09685.
- [31] Z. Liu et al., “SpinQuant: LLM Quantization with Learned Rotations,” arXiv:2405.16406, 2024.
- [32] E. Oja, “A Simplified Neuron Model as a Principal Component Analyzer,” *J. Mathematical Biology* 15, 267–273, 1982.
- [33] V. Joshi et al., “Accurate deep neural network inference using computational phase-change memory,” *Nature Communications* 11, 2473, 2020.
- [34] K. Vandoorne et al., “Experimental demonstration of reservoir computing on a silicon photonics chip,” *Nature Communications* 5, 3541, 2014.
- [35] G. Tanaka et al., “Recent advances in physical reservoir computing: A review,” *Neural Networks* 115, 100–123, 2019.
- [36] K. Y. Camsari, R. Faria, B. M. Sutton, S. Datta, “Stochastic p-bits for invertible logic,” *Phys. Rev. X* 7, 031014, 2017.
- [37] R. N. M. Watson et al., “CHERI: A Hybrid Capability-System Architecture for Scalable Software Compartmentalization,” *IEEE S&P*, 2015.
- [38] P. Kocher et al., “Spectre Attacks: Exploiting Speculative Execution,” *IEEE S&P*, 2019.